

ChE 312 – Transport Phenomena II –Spring 2025

Rubric Assessment Summary

The assessment of ABET Student Outcome 1 was conducted using the final exam as the primary direct assessment instrument. The exam was scored out of 100 points.

ABET Outcome 1

An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Performance Indicator (PI)	Level 3: Satisfactory >45	Level 2: Developing 35 to 45	Level 1: Unsatisfactory <35
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
6. The uses of appropriate resources needed to solve problems	Uses appropriate resources to locate information needed to solve problems	Uses limited resources to solve problems	Uses no resources to solve problems

FINAL EXAM TOTAL POINTS 100

Average Score	39.25
Standard dev.	19.95

RESULTS BY PERFORMANCE INDICATOR

Mapped SO	1
Average	1.85
Std Dev	0.81
S	5
D	7
U	8

Summary of Findings

The overall student performance on the final exam indicates **low attainment of course learning outcomes with significant variability**. The **average exam score was 39.25**, with a **standard deviation of 19.95**, demonstrating both weak overall performance and a wide spread in student achievement levels. This suggests that many students are struggling with fundamental concepts and problem-solving requirements.

For the mapped performance indicator, the **average rubric score was 1.85 (on a 3-point scale)** with a **standard deviation of 0.81**. This value is **below the attainment benchmark of 2.0**, indicating that **Student Outcome 1 is not attained** in this assessment cycle.

The distribution of performance further confirms this concern:

- **Satisfactory (S): 5 students**
- **Developing (D): 7 students**
- **Unsatisfactory (U): 8 students**

Interpretation

A majority of students (**15 out of 20, or 75%**) did not reach satisfactory performance, with a particularly high proportion in the unsatisfactory category. This indicates widespread difficulty in applying engineering principles to solve complex problems.

Overall, the results show that students are **not meeting expected competency levels for SO1**. The low average score and rubric attainment indicate deficiencies in students' ability to:

- Translate physical engineering problems into mathematical formulations
- Apply fundamental engineering principles correctly

- Execute multi-step problem-solving processes
- Interpret and validate engineering results

The relatively high standard deviation in both exam scores and rubric performance suggests **uneven preparation and inconsistent mastery across the class**, with only a small subset of students demonstrating strong proficiency.

These results indicate that current instructional and assessment strategies are **not sufficiently supporting student mastery of core problem-solving skills** required for success in the program.

Recommended Improvements

To improve student attainment of SO1, the following actions are recommended:

A. Strengthen Foundational Knowledge

- Reinforce prerequisite concepts (e.g., mathematics, physics, engineering fundamentals) at the beginning of the course.
- Provide structured review materials to support students with weak background preparation.

B. Increase Structured Practice and Feedback

- Introduce frequent **low-stakes assignments, quizzes, and guided problem sets**.
- Provide **timely, detailed feedback** to correct misconceptions early.

C. Improve Problem-Solving Instruction

- Use **step-by-step problem-solving frameworks** to guide students through complex engineering problems.
- Emphasize conceptual understanding before mathematical execution.
- Incorporate more worked examples demonstrating common exam-level problem types.

D. Enhance Student Engagement and Support

- Implement **active learning strategies** such as in-class problem solving and peer instruction.
- Offer **targeted tutoring, review sessions, and office hour support**, particularly for at-risk students.

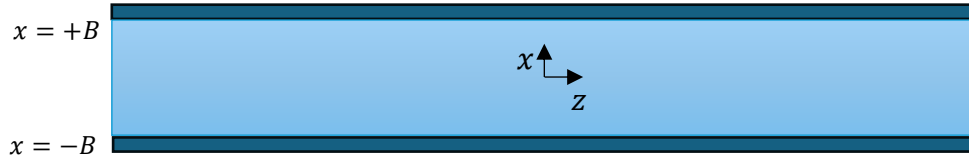
E. Review Course Alignment and Assessment Design

- Conduct a detailed review of alignment between **learning outcomes, instructional content, and exam difficulty**.
- Perform item-level analysis of exam questions to identify specific areas of student weakness.
- Refine rubric descriptors to ensure consistent and meaningful performance classification.

Conclusion

Based on the assessment results, **Student Outcome 1 is not attained** in this course offering. The data indicates significant gaps in student understanding and application of engineering problem-solving skills. The recommended continuous improvement actions

focus on strengthening foundational knowledge, increasing structured practice and feedback, enhancing instructional strategies, and improving alignment between teaching and assessment. These measures will be implemented and evaluated in the next course cycle to improve student performance and outcome attainment.



The above schematic and the reference frame apply to both problems.

1. Liquid B is laminarly flowing between two flat walls placed a distance $2B$ apart, with the velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

The solid walls are made of a species A that is slightly soluble in the liquid B . Species A diffuses into the liquid according to Fick's law for binary diffusion with a constant diffusivity D_{AB} , where it undergoes a chemical reaction according to a first order rate law $R_A = -k_1''' c_A$.

Answer the questions in terms of the following parameters (all assumed constant):

$v_{z,max}$: maximum velocity of the fluid in z -direction

B : half the distance between the two parallel walls

D_{AB} : Fick's law diffusivity of A in B

k_1''' : rate constant in units of inverse time.

- 1.1. By setting up a shell mole balance, derive a partial differential equation for the concentration of species A as a function of x and z . Assume the concentration of A is very small $x_A \ll 1$, and neglect the diffusive transport of A in z direction.

$$\int_V dV \frac{\partial}{\partial t} c_A = - \int_S dS \langle N_A | n \rangle + \int_V dV R_A$$

$$\frac{\partial N_{Ax}}{\partial x} + \frac{\partial N_{Az}}{\partial z} + k_1''' c_A = 0$$

$$\frac{\partial}{\partial x} \left(-D_{AB} \frac{\partial c_A}{\partial x} \right) + \frac{\partial}{\partial z} (c_A v_z) + k_1''' c_A = 0$$

$$-D_{AB} \frac{\partial^2 c_A}{\partial x^2} + v_{z,max} \left[1 - \left(\frac{x}{B} \right)^2 \right] \frac{\partial c_A}{\partial z} + k_1''' c_A = 0$$

- 1.2. Sufficiently far downstream, the concentration profile c_A is independent of z . Find this asymptotic concentration profile as a function of x , assuming the concentration of A at both walls is c_{A0} .

$$\frac{d^2 c_A}{dx^2} = \frac{k_1'''}{D_{AB}} c_A$$

$$c_A = C_1 \cosh \sqrt{\frac{k_1'''}{D_{AB}}} x + C_2 \sinh \sqrt{\frac{k_1'''}{D_{AB}}} x$$

$$c_A(\pm B) = c_{A0} \rightarrow C_2 = 0, C_1 = \frac{c_{A0}}{\cosh \sqrt{\frac{k_1'''}{D_{AB}}} B}$$

$$c_A = \frac{c_{A0}}{\cosh \sqrt{\frac{k_1'''}{D_{AB}}} B} \cosh \sqrt{\frac{k_1'''}{D_{AB}}} x$$

2. Very thick slurries and pastes sometimes move in a channel almost as a solid plug with a uniform velocity profile. Consider a fluid between two flat walls placed a distance $2B$ apart, with a uniform velocity profile $v_z = v_0$. For $z > 0$, heat is added to the fluid with a constant flux q_0 at both walls. The liquid enters the heated region of the slit at $z = 0$ with a uniform temperature T_0 .

Answer the questions in terms of the following parameters (all assumed constant):

- v_0 : velocity of the fluid in z -direction
 B : half the distance between the two parallel walls
 k : thermal conductivity of the fluid
 ρ : density of the fluid
 c_p : isobaric heat capacity per unit mass of the fluid
 q_0 : heat flux at the walls (assumed positive)

- 2.1. Write down a steady state macroscopic energy balance on the entire fluid between 0 and z to find the bulk average temperature as a function of z . Assume constant heat capacity and neglect the pressure dependence of enthalpy.

$$\int_V dV \frac{\partial}{\partial t} \rho \left(\frac{1}{2} v^2 + \phi + u \right)$$

$$= - \int_S dS \rho \left(\frac{1}{2} v^2 + \phi + u \right) \langle v|n \rangle - \int_S dS \langle q|n \rangle - \int_S dS \langle v|\tau|n \rangle - \int_S dS p \langle v|n \rangle$$

$$T_b(z) \equiv \frac{\int_A dS v_z T}{\int_A dS v_z}$$

Velocity and potential energy are independent of z and $h = u + \frac{p}{\rho}$,

$$0 = - \int_S dS \rho h \langle v|n \rangle - \int_S dS \langle q|n \rangle$$

$$0 = \int_A dS \rho [h(T_0) - h(T(x, z))] v_0 + 2q_0 z W$$

A is the flow cross-section and $\Delta h = c_p \Delta T$

$$0 = \int_A dS \rho c_p [T_0 - T(x, z)] v_0 + 2q_0 z W$$

$$0 = 2BW \rho c_p v_0 [T_0 - T_b(z)] + 2q_0 z W$$

$$T_b(z) = T_0 + \frac{q_0}{B \rho c_p v_0} z$$

- 2.2. Write down a steady state shell energy balance to obtain a partial differential equation (PDE) for the temperature of the liquid as a function of x and z .

$$0 = - \int_S dS \rho h \langle v | n \rangle - \int_S dS \langle q | n \rangle$$

Apply the above energy balance to a shell volume of $\Delta x \Delta z W$

$$0 = [\Delta x W \rho v_0 h]_z - [\Delta x W \rho v_0 h]_{z+\Delta z} + [\Delta z W q]_x - [\Delta z W q]_{x+\Delta x}$$

$$\rho v_0 \frac{\partial h}{\partial z} + \frac{\partial q}{\partial x} = 0$$

$$v_0 \rho c_p \frac{\partial T}{\partial z} - k \frac{\partial^2 T}{\partial x^2} = 0$$

For large z , the temperature profile assumes the following asymptotic solution:

$$T(x, z) = T_b(z) + \psi(x)$$

- 2.3. Derive an ordinary differential equation (ODE) for $\psi(x)$ by replacing the asymptotic solution in the PDE derived in the previous part. Solve the ODE and evaluate the integration constants.

$$\frac{d^2 \psi}{dx^2} = \frac{q_0}{kB}$$

$$\psi = \frac{q_0}{2kB} x^2 + c_1 x + c_2$$

$$\text{B.C. 1: } -k \frac{d\psi}{dx} \Big|_{x=B} = -q_0 \text{ or } -k \frac{d\psi}{dx} \Big|_{x=-B} = q_0 \rightarrow c_1 = 0$$

$$\text{B.C. 2: } \int_0^B dx \left(\frac{q_0}{2kB} x^2 + c_2 \right) = 0 \rightarrow c_2 = -\frac{q_0 B}{6k}$$

- 2.4. Using the asymptotic temperature profile derived above find the Nusselt number.

$$Nu = \frac{4hB}{k}$$

$$h[T(B,z)-T_b(z)]=q_0$$

$$T(x,z)=T_0+\frac{q_0}{B\rho c_p v_0}z+\frac{q_0}{2kB}x^2-\frac{q_0B}{6k}$$

$$T(B,z)=T_0+\frac{q_0}{B\rho c_p v_0}z+\frac{q_0B}{3k}$$

$$h=\frac{q_0}{T(B,z)-T_b(z)}=\frac{3k}{B}$$

$$Nu=\frac{4\left(\frac{3k}{B}\right)B}{k}=12$$